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A hybrid augmented ant colony optimization for the multi-trip capacitated arc routing problem under fuzzy demands for urban solid waste management

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Abstract

Nowadays, urban solid waste management is one of the most crucial activities in municipalities and their affiliated organizations. It includes the processes of collection, transportation and disposal. These major operations require a large amount of resources and investments, which will always be subject to limitations. In this paper, a chance-constrained programming model based on fuzzy credibility theory is proposed for the multi-trip capacitated arc routing problem to cope with the uncertain nature of waste amount generated in urban areas with the aim of total cost minimization. To deal with the complexity of the problem and solve it efficiently, a hybrid augmented ant colony optimization algorithm is developed based on an improved max–min ant system with an innovative probability function and a simulated annealing algorithm. The performance of hybrid augmented ant colony optimization is enhanced by using the Taguchi parameter design method to adjust the parameters' values optimally. The overall efficiency of the algorithm is evaluated against other similar algorithms using well-known benchmarks. Finally, the applicability of the suggested methodology is tested on a real case study with a sensitivity analysis to evolve the managerial insights and decision aids.

Keywords

Urban solid waste management, multi-trip capacitated arc routing problem, fuzzy credibility theory, hybrid augmented ant colony optimization, improved max–min ant system

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Introduction

Uncontrolled urban expansion and the massive increase in urban populations have led to a vast amount of consumption and different types of waste generation over the past years. In 2012, various cities in the world generated 1.3 billion tons of solid waste, equivalent to 1.2 kg per person per day (Statista, 2018). With this upcoming trend and rising population and urbanization rates, it is expected that the annual waste generation rate will reach 2.2 billion tons by 2025 (Statista, 2018). This problem is very critical even in developed countries, for example, in Western Europe; despite the existence of a successful education system and raising awareness, the annual per capita generation of solid waste is over 600 kg (Gapminder, 2017). The daily waste generation of Iran is 20 million tons, which is three times greater than the global standard announced by the World Health Organization (Qarehgozlou, 2017). This amount of waste generation definitely leads to an increase in the necessary funds for collection and transportation operations, which contains 75%–80% of urban solid waste management (USWM) costs (Tirkolaee et al., 2018a). The assigned budgets to municipalities and their local organizations for USWM are over 1 billion dollars in Malaysia including about 70%–80% of the revenue of these organizations (Budhiarta et al., 2012). This amount has been reported at 382.2 million dollars just for Toronto in Canada (Toronto City Council, 2018). These factors illustrate why USWM has been one of the most important concerns of human societies in recent decades. Urban areas generate the highest amount of waste, and an efficient system to collect and dispose of waste is needed. It is very difficult and costly to set up and maintain a proper USWM system. In fact, USWM processes are the requirements of each city and lack of attention to these processes leads to irreparable effects on the residents, environment and even residents of villages on the outskirts (Kirkeby et al., 2006).

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On the other hand, the exact amount of generated waste is always unknown in each urban area due to its uncertain nature. Managers and decision-makers are interested to deal with the lowest possible amount of cost. However, each area may have a lower and an upper bound for the amount of waste according to its minimum and maximum values in a given time period. Therefore, an optimization system is needed to assess these associated costs, considering the fluctuations of the generated waste. To this end, this research investigates the capacitated arc routing problem (CARP) for USWM under uncertain demand. CARP is a classical combinatorial optimization problem that aims at determining optimal routes for vehicles to cover a set of edges with demand, which is called required edges, within a graph network considering operational limitations such as vehicles' capacity constraint (Golden et al., 2002). Furthermore, minimizing the number of required vehicles to cover all the required edges is regarded along with total traversing cost minimization in CARPs. Despite the large application of this problem in practice, just a few research works have investigated it in the literature.

The remainder of the current paper is organized as follows: the second section provides a literature review of the research; the proposed fuzzy credibility theory is discussed in the third section in detail; the fourth section provides the problem definition and the fuzzy mathematical model; the fifth section describes the suggested hybrid augmented ant colony optimization (HA-ACO), and the obtained comparison results on the instance problems and well-known problems are represented in the sixth section; the seventh section presents a real case study problem in Iran; the conclusion and outlook of the research are detailed in the eighth section.

Literature review

To the best of our knowledge, Beltrami and Bodin (1974) were the first researchers who presented a classic vehicle routing problem (VRP) for urban waste collection application in New York and Washington municipalities considering various kinds of vehicles. They employed and improved the saving algorithm introduced by Clarke and Wright (1964). Finally, they could achieve high benefits by implementing their suggested methodology in both cities. After them, many researchers have worked on VRPs and location-routing problems for USWM considering different characteristics such as hazardous waste and multiple waste products (Aydemir-Karadag, 2018; Hauge et al., 2014; Louati, 2016; Paredes-Belmar et al., 2017; Rabbani et al., 2018; Ramos et al., 2018; Tung and Pinnoi, 2000; Yilmaz et al., 2014). Furthermore, the VRP has been studied widely in the literature for various real-world applications (Mirmohammadi et al., 2017; Tirkolaee et al., 2017, 2019), but CARP has been almost neglected despite its effective practical aspects, particularly in USWM (Tirkolaee et al., 2018a). Moreover, some researchers have worked on the application of a Geographic Information System to construct optimal routing for USWM (Farahbakhsh and Forghani, 2019; Sanjeevi and Shahabudeen, 2016).

In the following, different variants of CARP for USWM are reviewed. Ghiani et al. (2001) introduced CARP with intermediate facilities (CARP-IF), which can be implemented for urban waste collection. They offered two lower bounds and two upper bounds based on the rural postman problem and CARP for their proposed problem. In another research, Ghiani et al. (2010) developed the CARP-IF with capacity and distance constraints using a novel ant colony optimization (ACO). They could improve some best-known solutions provided by other heuristics. A CARP-IF for two types of applications was suggested by Polacek et al. (2008). They considered a subset of intermediate equipment to load (direct service, for example, spreading salt before/after the snow) or unload vehicles (indirect service, for example, waste collection).

An evaluation of CARPs for USWM in developing countries was performed by Amponsah and Salhi (2004). They declared that the waste collection problem will be more decisive for big cities in developing countries due to the high temperature of the weather. They employed a constructive heuristic algorithm to solve the problem considering the environmental and economic aspects. Del Pia and Filippi (2006) proposed a novel CARP with two different types of vehicles where just the first type of vehicles (as the larger vehicles) could unload the collected waste at the disposal site, and the second type of vehicles (as the smaller vehicles) could unload at the first type of vehicles. They could create significant savings in terms of total traversing cost by applying their suggested problem. Mourão et al. (2009) proposed three heuristic methods in order to solve the sectoring arc routing problem (SARP) for USWM. In their suggested SARP, a street network was divided into several sectors, then a set of vehicle trips was assigned to each district to minimize the total traversing time. Cortinhal et al. (2016) applied several local search heuristics for SARP in a household waste-collection issue. They planned the sectors to concurrently minimize the total routing time, the workload time imbalance and the number of connected components. They employed the normalized weighted sum approach to have a single-objective model and tested their heuristics on random and real-life instances.

Recently, a two-phased hybrid local search algorithm was proposed for the periodic CARP (PCARP) was suggested by Chen and Hao (2018) for urban waste-collection application. They studied the minimization of the total traversing cost and total number of vehicles simultaneously. They could improve 44 best-found upper bounds out of 63 benchmarks. Tirkolaee et al. (2018a) developed an improved ACO to solve the multi-trip CARP for the urban waste-collection problem. They addressed the problem using a mathematical model and could implement their suggested algorithm on benchmarks and randomly generated instance problems to assess its efficiency. Arakaki and Usberti (2019) proposed an efficiency-based path-scanning heuristic algorithm for CARP, which can be implemented in USWM. They could outperform all previous path-scanning heuristics in the literature. A multi-compartment CARP with intermediate facilities was presented by Mofid-Nakhaee and Barzinpour

(2019) for solid waste collection. They developed two algorithms of a hybrid adaptive large neighborhood search and whale optimization algorithm to cope with the complexity of the problem. Finally, they conducted a real case study in Tehran and concluded that multi-compartment vehicles are more efficient than single-compartment vehicles.

Here, the main research conducted for studying the effects of uncertainty is reviewed as follows. Fleury et al. (2004, 2005) introduced the stochastic arc routing problem in which some parameters such as customers' demands and traversing times are random. Laporte et al. (2010) investigated a CARP with stochastic demands considering service failures due to vehicle capacity constraint violation. They applied a neighborhood search heuristic algorithm to solve the problem.

Babaee Tirkolaee et al. (2016) introduced the robust CARP for urban waste-collection and designed a hybrid simulated annealing (SA) algorithm to solve it. They could prove the efficiency of the algorithm in comparison with the proposed mathematical model. Tang et al. (2017) offered a scalable technique to solve CARP for the urban waste-collection problem. The procedure of their solution method was to hierarchically decompose the tasks into sub-tasks and solve them recursively using a local search technique. Recently, Tirkolaee et al. (2018b) designed a novel formulation for the robust PCARP on USWM considering drivers and crew's working time and uncertain demand. They proposed an efficient SA algorithm to solve the problem and evaluated its performance compared to the mathematical model.

As can be seen, there is scant research conducted in the literature of CARP to study the impacts of uncertainty. For this reason, our investigation focuses on the multi-trip CARP under fuzzy demands. The main contributions of this paper are as follows, trying to suggest a comprehensive methodology to be implemented in related organizations such as municipalities:

1. studying the uncertainty impacts as a chance-constrained programming model based on the fuzzy credibility theory in the multi-trip CARP;
2. introducing a HA-ACO based on an improved max–min ant system (IMMAS) with a novel probability function and SA algorithm to solve the proposed model, which is the extension of ACO proposed by Tirkolaee et al. (2018a);
3. sdjusting the optimal values of the algorithm's parameters using the Taguchi design technique; and
4. implementing the proposed methodology on a real case study with effective sensitivity analysis in order to evaluate its applicability.

Fuzzy set theory and fuzzy credibility criterion

When parameters of a problem fluctuate according to the real-world conditions and lead to the uncertainty of the problem, and also when no historical data are available for these parameters, estimating the probability distribution is almost impossible.

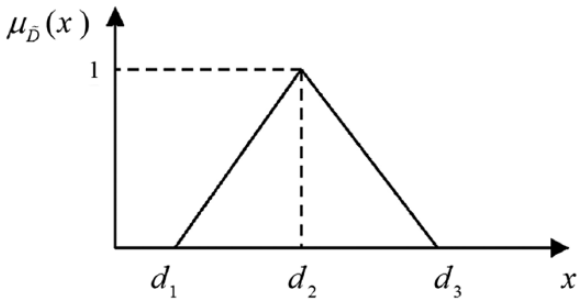


Figure 1. Membership function of a triangular fuzzy number.

Therefore, stochastic programming can hardly be applied efficiently to our case. In this situation, fuzzy set theory can assist the uncertainty modeling effectively (Dubois and Prade, 2015). A classic set is generally defined as a set of its members. $\tilde{D} = (d_1, d_2, d_3)$ Each datum may be a member of this set or not. As a result, the membership or non-membership of each datum can be clearly determined. However, the membership is not associated with certainty in some sets. To deal with such sets, fuzzy set theory is employed. It was first introduced by Zadeh (1965), and was developed for a variety of problems. Moreover, fuzzy variables were first introduced by Kaufmann (1975), and then by Zadeh (1975) and Nahmias (1978). Years later, Liu (2004) found out the fuzzy credibility theory in his research such that it has been studied very recently by many researchers (Jiang et al., 2018; Mehrjerdi and Nadizadeh, 2013).

In this section, the basic concepts of the fuzzy chance-constrained programming based on the credibility theory are introduced. Initially, the principles of the possibility criterion are discussed as the basis of the credibility theory. The credibility of a fuzzy event is defined as the average of its possibility and necessity (Nadizadeh and Nasab, 2014). A fuzzy event may lead to failure, even though the possibility of its occurrence is one. On the other hand, it may happen that even its necessity is zero. Therefore, the credibility criterion uses the combination of these two functions and plays the role of an event probability in a fuzzy environment.

In this research, we consider a triangular fuzzy number for the fuzzy variable $\tilde{D} = (d_1, d_2, d_3)$ so that its membership function is represented by Figure 1 and equation (1)

$$\mu_{\tilde{D}}(x) = \begin{cases} \dfrac{(x - d_1)}{(d_2 - d_1)}, \ if \ d_1 \le r < d_2, \\ 1, if \ x = d_2, \\ \dfrac{(d_3 - x)}{(d_3 - d_2)}, if \ d_2 < r \le d_3, \\ 0, otherwise. \end{cases} \quad (1)$$

Thus, the dispatcher of the vehicles or analyst of the problem may empirically make a decision based on his/her experience and insight or available (historical) data. In this way, the demand of a required edge is not less than d_1 (left boundary) and more than d_3 (right boundary), and d_2 (middle boundary) is the most

reasonable value. In this research, demands of the required edges are taken into account as triangular fuzzy numbers $\tilde{D}=(d_1,d_2,d_3)$.

Now, if $\mu_{\tilde{D}}(x)$ is the membership function of the triangular fuzzy demand $\tilde{D}$, then the possibility, necessity and credibility functions of fuzzy event $\{\tilde{D}\geq r\}$ are defined by equations (2)–(4), respectively (Liu, 2004)

$$Pos\left\{\tilde{D}\geq r\right\}=\begin{cases}1, if\ r<d_2,\\ \dfrac{d_3-r}{d_3-d_2}, if\ d_2\leq r\leq d_3,\\ 0, if\ r>d_3,\end{cases} \tag{2}$$

$$Nec\left\{\tilde{D}\geq r\right\}=\begin{cases}1, if\ r<d_1,\\ \dfrac{d_2-r}{d_2-d_1}, if\ d_1\leq r\leq d_2,\\ 0, if\ r>d_2,\end{cases} \tag{3}$$

$$Cr\left\{\tilde{D}\geq r\right\}=\begin{cases}1, & if\ r<d_1,\\ \dfrac{2d_2-d_1-r}{2(d_2-d_1)}, & if\ d_1\leq r\leq d_2,\\ \dfrac{d_3-r}{2(d_3-d_2)}, & if\ d_2<r\leq d_3,\\ 0, & if\ r>d_3.\end{cases} \tag{4}$$

where r is a random value for the demand parameter to be compared with d_1, d_2, and d_3.

Problem definition

According to the public health and environmental principles, wastes as sources of contamination should be disposed of as soon as possible, otherwise, the transmission of diseases, bad smells and their leachate leakage cause severe pollution to the environment. For these reasons, disposal facilities should be located outside of the urban areas in USWM systems, while vehicles' depots are located in urban areas due to the economic aspects. Therefore, in this study, a given undirected graph $G = (V, E)$ is addressed to formulate the problem, where V and E denote the set of nodes and the set of edges, respectively. In this graph, nodes 1 and n (first node and last node) represent the depot and disposal facility, and each edge in the network is denoted by $e(i,j)\in E: i\neq j$. Furthermore, E_R represents the set of the required edges. It should be noted that each undirected edge $e(i,j)\in E$ comprises two directed arcs (*1.* arc (i,j) and *2.* arc (j,i)), and the vehicles should serve just one of these arcs related to the required edges. The objective of the problem is to find the optimal number of vehicles and the optimal routes to minimize the total cost. The main assumptions of the suggested mixed-integer linear programming model are as follows:

(*A1*) Demands of the required edges are defined as triangular fuzzy numbers.

(*A2*) Vehicles depot and disposal facilities are located in different nodes.

(*A3*) Required edges must be served just by one vehicle.

(*A4*) Vehicles may have multiple trips based on their maximum available time. The first trip starts from the depot and ends at the disposal facility. The next potential trips start from the disposal facility and end at there again.

(*A5*) Vehicles are heterogneous.

Mathematical model

The proposed mathematical model is adapted from Tirkolaee et al. (2018a) to study the uncertainty. The main differences are that we consider heterogenous vehicles and incorporate the uncertainty of demand parameter in the model. The first effect of the uncertainty appears in the vehicles' capacity constraint, and accordingly, the results may change under different uncertain conditions. In the following, the suggested fuzzy model is described:

Sets:	
$i, j, h \in V$	Indexes of nodes
$k \in K$	Vehicle index
$p \in P$	Indexes of vehicles' trips
E	Set of edges
E_R	Set of required edges
S	Optional subset of edges
$V[S]$	Set of nodes in S
$Z^+ = \{1, 2, 3, \ldots\}$	Set of positive integers
Parameters:	
t_{ij} (min)	Traversing time of $e(i, j)$
c_{ij} ($)	Traversing cost of $e(i, j)$
$\tilde{d}_{ij}$ (kg)	Fuzzy demand of required edge $e(i, j)$
cv_k ($)	Usage cost of kth vehicle
T_{max} (hr)	Maximum available usage time for vehicles
W_k (kg)	Capacity of kth vehicle
M	An arbitrary big number

Decision variables

$$x_{ijkp}=\begin{cases}r, & \text{if } e(i,j)\in E \text{ is traversed}\\ & r \text{ times } (r\in Z^+) \text{ by } k^{\text{th}} \text{ vehicle in } p^{\text{th}} \text{ trip},\\ 0, & \text{otherwise},\end{cases} \tag{5}$$

$$y_{ijkp}=\begin{cases}1, & \text{if } e(i,j)\in E \text{ is served by}\\ & k^{\text{th}} \text{ vehicle in } p^{\text{th}} \text{ trip},\\ 0, & \text{otherwise},\end{cases} \tag{6}$$

$$u_k=\begin{cases}1, & \text{if } k^{\text{th}} \text{ vehicle is used},\\ 0, & \text{otherwise}.\end{cases} \tag{7}$$

Now, the objective function and constraints are presented as follows

$$\text{minimize } Z=\sum_{e(i,j)\in E}\sum_{p\in P}\sum_{k\in K}c_{ij}\,x_{ijkp}+\sum_{k\in K}cv\,u_k \tag{8}$$

subject to

$$\sum_{j\in V}x_{jikp}=\sum_{j\in V}x_{ijkp}\qquad \forall i\in V,\forall k\in K,\forall p\in P, \tag{9}$$

$$\sum_{p\in P}\sum_{k\in K}(y_{ijkp}+y_{jikp})=1\qquad \forall e(i,j)\text{ or }e(j,i)\in E_R, \tag{10}$$

$$Cr\left(\sum_{e(i,j)\in E_s}\tilde{d}_{ij}\,y_{ijkp}\le W_k\right)\ge Cr^*\qquad \forall k\in K,\forall p\in P, \tag{11}$$

$$y_{ijkp}\le x_{ijkp}\qquad \forall e(i,j)\in E,\forall k\in K,\forall p\in P, \tag{12}$$

$$\sum_{p\in P}\sum_{e(i,j)\in E}x_{ijkp}\le M\,u_k\qquad \forall k\in K, \tag{13}$$

$$\sum_{p\in P}\sum_{e(i,j)\in E}t_{ij}x_{ijkp}\le T_{\max}\qquad \forall k\in K, \tag{14}$$

$$\sum_{\substack{e(1,j)\in E\\ j\in V\setminus\{1,n\}}}x_{1jkp}=u_k\qquad \forall k\in K,\ \forall p=1, \tag{15}$$

$$\sum_{\substack{e(j,n)\in E\\ j\in V\setminus\{1,n\}}}x_{jnkp}=u_k\qquad \forall k\in K,\ \forall p=1, \tag{16}$$

$$\sum_{\substack{e(n,j)\in E\\ j\in V\setminus\{1,n\}}}x_{njkp}\le u_k\qquad \forall k\in K,\forall p\in P\setminus\{1\}, \tag{17}$$

$$\sum_{\substack{e(j,n)\in E\\ j\in V\setminus\{1,n\}}}x_{jnkp}\le u_k\qquad \forall k\in K,\ \forall p\in P\setminus\{1\}, \tag{18}$$

$$\sum_{e(j,h)\in S}x_{jhkp}\le M\sum_{i\notin V[S],\,j\in V[S]}x_{ijkp}\qquad \forall S\subseteq E,\ \{1,n\}\notin V[\mathrm{S}],\forall k\in K,\ \forall p\in P, \tag{19}$$

$$x_{ijkp}\in Z^+,y_{ijkp}\in\{0,1\},u_k\in\{0,1\}\qquad \forall e(i,j)\in E,\ \forall k\in K,\forall p\in P. \tag{20}$$

The objective function equation (8) aims at minimizing the total cost including total traversing cost and total vehicles' usage cost. Equation (9) guarantees route continuity constructed by each vehicle. Equation (10) ensures that each required edge should be assigned to a vehicle by serving one of its constituting twin arcs. Equation (11) states the capacity constraint of each vehicle for a specific confidence level. Equation (12) reflects that each vehicle may traverse an edge without serving it. Equation (13) ensures that each vehicle may be used when its usage cost is paid. Equation (14) represents the maximum available time for each vehicle. Equations (15) and (16) guarantee that the first trip starts from the depot and ends at the disposal facility for each vehicle. Equations (17) and (18) state that each vehicle starts its possible next trips (except its first trip) from the disposal facility such that each trip must end at the disposal facility again. Equation (19) expresses the sub-tour elimination. Finally, the types of the decision variables are represented by equation (20).

The effect of credibility level on solution quality

The demands of the required edges are estimated based upon the credibility level, and then routing is conducted with respect to these estimated demands. The preference index is described by $Cr\in[0,1]$, which stands for the extent of our preference to send the vehicle to the next required edge after serving the current required edge based on equation (11). For $Cr=0$, we surely know that the vehicle should go back to the disposal facility, and for $Cr=1$ we are quite sure that the vehicle should serve the next required edge.

We define the dispatcher preference index as $Cr^*\in[0,1]$, where Cr^* represents the dispatcher's attitude toward risk. When the dispatcher is a risk lover, he/she will accept lower values for Cr^*, which reflects that the dispatcher is willing to use the vehicle's available capacity to the maximum extent possible, although the failure probability to fulfill the planned service increases when the vehicle arrives at the next required edge such that it may not be able to serve due to its low available capacity. On the other hand, when the dispatcher chooses a higher value for Cr^*, he/she ensures that the available capacity of the vehicle is greater than the demand of the next required edge with a high probability, and it shows that the dispatcher is risk-averse. Hence, finding the optimal value of Cr plays a determinant role for the real value of the objective function. As the initial selection of the required edges demands is based on the credibility of the estimations, vehicles may encounter the demands that are different from the predetermined ones. Selecting low values of Cr would lead to estimate the demands at their minimum possible values. Although this leads to a reduction in the costs of the constructed routes, the possibility of failure in the routes will increase because the estimated values of demands may be less than their actual values. It should be noted that route failure means that the vehicle has to move back to the disposal facility and empty the collected waste in order to serve the remaining required edges, if possible (based on its remaining usage time). On the other hand, choosing a high value of Cr leads to consideration of an upper bound for each demand value so that in this case, the generated routes remain feasible and the possibility of route failure becomes zero. However, due to the overestimation of the demands, the vehicle capacity is not used properly and consequently, the fixed routing costs may increase.

In the following, a Monte Carlo simulation tool is described to be implemented with the aim of finding the best credibility value

Step 1: Repeat steps 2 to 5 for $Cr = 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9$ and 1,

Step 2: Determine the demands of the required edges for the selected level of Cr, and find the optimal solution of the problem accordingly. The demands of required edges are specified using the following sub-steps algorithm.

Sub-step 1: Set i=1.

Sub-step 2: Generate a random number m_i between the lower and upper bounds of fuzzy demand for $e(i, j)$, determine its membership value (f) based on the demand membership function of $e(i, j)$.

Sub-step 3: Generate a random number r in [0, 1].

Sub-step 4: If $r \leq f$, consider m_i as the real demand value of $e(i, j)$ and go to Sub-step 5; otherwise, go to Sub-step 2.

Sub-step 5: Set i:=i+1, if $i \leq n$, go to Sub-step 2; otherwise, go to Sub-step 6.

Sub-step 6: Finish.

Step 3: Calculate the costs of optimal routes for a given Cr according to the simulated demand of required edges. These costs are regarded as the costs of the occurred failures.

Step 4: Run Steps 2 and 3 for 1000 times and calculate the average costs as the additional costs of route failure.

Step 5: Report the value of Cr with the least average of the objective function as the optimal Cr.

Step 6: Stop.

Figure 2. Monte Carlo simulation algorithm.

such that the required edges' demands can be estimated carefully according to the given credibility values.

Determining the optimal credibility level

In our methodology, the decisions related to dispatching vehicles are made as follows. When the condition of $Cr \geq Cr^*$ is satisfied, then the vehicle should be sent to serve the next required edge. Otherwise, the vehicle should go back to the disposal facility, and then start its next trip (or go to the depot due to its usage time limitation and another vehicle should start its trip) to serve the next required edge. The above process will not be terminated until all the required edges are covered. Moreover, the optimal routes of the vehicles are determined in advance by implementing the proposed solution technique in the planning phase. However, the actual demand value of a required edge is only known once the vehicle starts serving the required edge. Therefore, due to the uncertainty of demand, a vehicle may not be able to serve a required edge when it arrives there. This is because of its insufficient remaining capacity when the vehicle follows the planned routes. Therefore, it is assumed in such a situation that the vehicle moves to the disposal facility and unloads what it has already collected. After that, it goes back to the operational area to serve the required edge where it had a "failure" and follows its services based on the remainder of the planned routes. Accordingly, the additional traversed distance is imposed due to this route failure.

In order to find the optimal value of Cr and additional distances traveled, the Monte Carlo simulation algorithm is implemented (cf. Figure 2).

By increasing Cr from zero to one, the value of the planned demands decreases; thus, it can be expected that the planned objective function decreases, too. However, the number of route failures also increases by increasing Cr. On the other hand, since the total objective function is the sum of the planned objective function and the cost of route failure, it can be claimed that the total objective function is minimized for a certain value of Cr.

Solution technique

Since the proposed problem is categorized as NP-hard (Santos et al., 2010), it cannot be solved efficiently by exact methods within a reasonable computational time. This research develops a metaheuristic solution method, namely, HA-ACO based on IMMAS and SA algorithm which is the extension of the suggested algorithm by Tirkolaee et al. (2018a).

The main aim of SA implementations in this study is to increase the quality of the initial solutions which directly influence the initial values of the pheromones in IMMAS. Furthermore, the effect of the demand parameter has been incorporated in the probability function of IMMAS as another improvement. This forces the vehicles to cover the high-demand required edges before the low-demand ones.

The ACO was first introduced by Dorigo (1992). He could present an ant system (AS) by being inspired by the social life of ants that he studied in his PhD thesis. After the introduction of AS, many researchers have worked to implement and develop it to solve different optimization problems such as the traveling salesman problem, assignment problems and routing problems that belong to the most applicable problems. The AS algorithm is the

Table 1. Improved max–min ant system parameters.

Parameter	Description
P_{ij}	Probability of traversing edge e(*i,j*)
τ_{ij}	Value of pheromone on edge e(*i,j*)
τ_{ij}^{new}	Update value of pheromone on edge e(*i,j*)
ρ	Pheromone evaporation rate
$0 < \delta < 1$	Pheromone deviation value from the upper bound of the pheromone trail
SP(*i, j*)	Shortest path between the nodes *i* and *j*
η_{ij}	$\frac{1}{SP(i,j)}$
A	Pheromone impact factor
B	Shortest path impact factor
C	Demand impact factor
$\Delta\tau_{ij}^{t}$	Pheromone deposited on edge e(*i,j*) by ants in t^{th} iteration
$\Delta\tau_{ij}^{best}$	Pheromone deposited on edge e(*i,j*) by the best ant until
$f(s_r)$	Objective value of rth solution in the population
$f(s_{best})$	Best objective value found in a given iteration
τ_{max}	Upper bound of pheromone can be deposited on each edge
K'	A constant value
μ_r	Weight of rth solution
n	Number of network nodes
N_i	Set of nodes in the neighborhood of node *i*
R	Set of solutions

first variant of ACO and the other well-known variants include elite ant system, rank ant system and max-min ant system (MMAS). These variants differ in the description of the probability function and the mechanism of pheromone deposition and evaporation.

The MMAS was suggested by Stützle and Hoos (2000) by applying major modifications in the AS algorithm. The most remarkable modification is that MMAS takes into account a lower and upper bound $[\tau_{\min}, \tau_{\max}]$ for the value of pheromone deposition in order to prevent an excessive concentration of the pheromones. Our proposed HA-ACO employs an IMMAS and utilizes the SA algorithm to augment the performance of IMMAS. IMMAS uses an additional mechanism of pheromone trail smoothing (PTS) to prevent precocious convergence (being trapped in local optimum solutions) and to provide more efficient search space (Stützle and Hoos, 2000). Table 1 describes the notations related to IMMAS, and the main equations of the suggested IMMAS are presented as follows

$$\mu_r = \frac{1}{f(s_r)} \quad \forall r \in R, \tag{21}$$

$$\tau_{ij}^{new} = \tau_{ij}(1-\rho) + \Delta\tau_{ij}^{best} + \sum_{r=1}^{R} \Delta\tau_{ij}^{r},\ \Delta\tau_{ij}^{r} = \frac{1}{f(s_r)},\ \Delta\tau_{ij}^{best} = \frac{1}{f(s_{best})} \quad \forall e(i,j) \in E, \forall r \in R, \tag{22}$$

$$\left[\frac{\tau_{Max}}{K'}, \tau_{Max}\right], \tau_{Max} = \frac{1}{\rho f(s_{best})},\ K' = \frac{(n-2)\sqrt[n]{0.05}}{2\left(1-\sqrt[n]{0.05}\right)}, \tag{23}$$

$$P_{ij} = \frac{\tau_{ij}^{A} \times \eta_{ij}^{B} \times d_{ij}^{C}}{\sum_{j} \tau_{ij}^{A} \times \eta_{ij}^{B} \times d_{ij}^{C}} \quad j \in N_i,\ e(i,j) \in E, \tag{24}$$

$$\tau_{ij}^{*}(\text{iteration}) = \tau_{ij}(\text{iteration}) + \delta(\tau_{\max}(\text{iteration}) - \tau_{ij}(\text{iteration})) \quad \forall e(i,j) \in E. \tag{25}$$

Equation (21) calculates the weight of rth solution where $f(s_r)$ is the objective function value of rth solution. Equation (22) represents the calculations of the pheromone trail updating. Equation (23) restricts the value of pheromone deposition on each edge based on the proposed mechanism in MMAS. The probability of traversing the remaining edges is shown in equation (24). Equation (25) reflects the PTS mechanism. According to PTS, the pheromone trail increases proportional to its deviation from the upper bound. The overall mechanism of the offered HA-ACO is displayed in Figure 3. As can be seen in Figure 3, HA-ACO employs a SA algorithm to improve the quality of the initial solutions which will affect the initial value of the pheromones; then the quality of the solutions constructed by ants will increase.

SA algorithm

After organizing the initial solutions and selecting the superior solutions, the SA algorithm is implemented to augment the quality of these selected solutions. The SA algorithm has shown a very successful performance to solve NP-hard problems in the literature (Glover and Kochenberger, 2006; Tirkolaee et al., 2018b). The general framework of the algorithm is such that it begins with initial solutions and improves them step by step. A neighborhood structure is followed for the initial solution. If the objective value of the generated neighbor is better than the objective function of the current solution, the solution will be replaced by the neighbor. Otherwise, the difference between the objectives of the neighbor solution and initial solution (Δ) is calculated. Then, a random number in [0, 1] will be produced and compared to the probability value of the algorithm, which is $e^{-\Delta/KT}$. If the generated random number is less than the probability value, the worse solution will be taken into account.

Initial solutions are generated using the heuristic method based on the greedy randomized adaptive search procedure suggested by Tirkolaee et al. (2018a). The number of initial solutions is 200. Then, the following local search methods are implemented to seek for a solution in the neighborhood in each iteration:

1. two different vehicles in a solution are selected randomly and exchanged with each other;
2. a required edge in a trip is exchanged with another required edge on that trip;

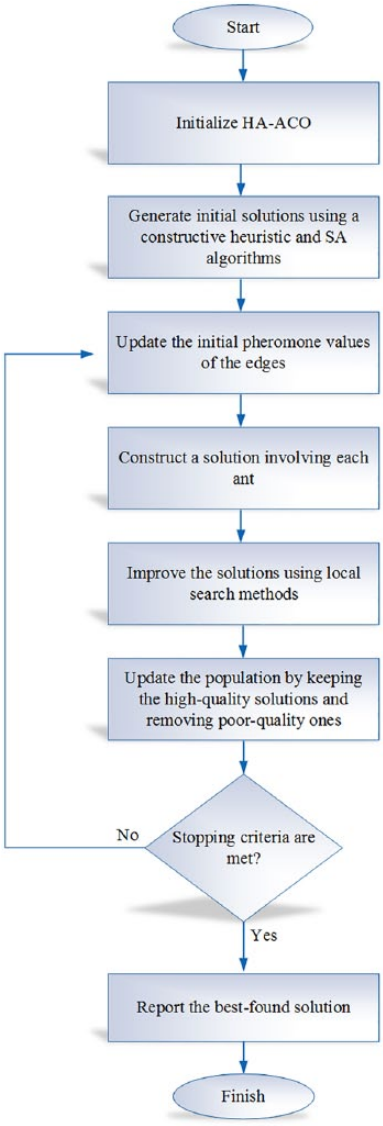


Figure 3. The flowchart of hybrid augmented ant colony optimization algorithm.

3. a required edge from a trip is exchanged with another edge from other trips by taking into account the vehicle's maximum available time and capacity limitation;
4. two required edges are replaced concurrently within a trip and between two trips; and
5. 2-opt (Beullens et al., 2003; Lacomme et al., 2004) and 3-opt (Maniezzo, 2004) operators are employed.

The parameters of SA are described in Table 2. Moreover, Figure 4 gives an overview of the SA algorithm as its suggested pseudo-code.

Parameters adjustment

As it is obvious, HA-ACO has several parameters which can affect the performance and final result. There are two well-known approaches to set the parameters (Taguchi et al., 2005): analysis of variance; and signal to noise (S/N) ratio. The S/N ratio indicates the dispersion around a given value. In other words, it shows how a solution will change among several experiments. Moreover, the S/N ratio suggested by the Taguchi method is applied to minimize the dispersion of objectives. It indicates noise factors along with controllable parameters (Taguchi et al., 2005). Tables 3 and 4 give the information about the parameters of IMMAS and SA with three levels of values, respectively. The full factorial plan requires $3^6 = 729$ experiments for IMMAS where six parameters may take three different values. As it is obvious, this experimental plan is not time- and cost-effective due to the high volume of computations. For this reason, the Taguchi design approach proposes a table related to an L^{27} array, which is implemented in Minitab statistical software. Therefore, the optimal values of HA-ACO's parameters are finally obtained by analyzing the mean of S/N ratios (cf. Table 5).

Table 2. Simulated annealing algorithm parameters.

Parameters	Description
M	Number of iterations at each temperature
T_0	Initial temperature
α	Cooling rate of temperature
T_{end}	Final temperature
K	Boltzmann constant

Begin
Choose an initail solution ($S1$)
Choose an initial temperature (T_0)
Repeat
  While (M<m)
    S2=Generate a neighbor of the solution $S1$
    Delta=Objective ($S1$) – Objective ($S2$)
    If Delta>0 Then
    S1:=S2
    Else if exp (Δ/KT) > Random (0, 1) Then
    $S1$:=$S2$
    End if
    M:=M+1
  End while
    T=αT
Until $T < T_{end}$

Figure 4. Pseudo-code of simulated annealing algorithm (Tirkolaee et al., 2018b).

Computational results

This section includes the evaluation of the proposed HA-ACO compared with the similar previously proposed algorithm in the literature. To this end, HA-ACO is tested on the randomly generated problems and benchmarks adapted from Tirkolaee et al. (2018a). The algorithm is implemented on a laptop with the system information of Intel® Core™ i7-4720HQ CPU @ 2.60 GHz, a RAM of 12.00 GB and 64-bit Operating System.

Table 3. Different levels defined for improved max–min ant system parameters.

Parameters	Level		
	1	2	3
Total iteration	100	150	200
Number of ants	5	10	15
A	1	2	3
B	1	3	5
ρ	0.1	0.2	0.25
δ	0.25	0.5	0.75

Table 4. Different levels defined for simulated annealing algorithm parameters.

Parameters	Level		
	1	2	3
M	1	2	5
T_0	150	200	300
α	0.8	0.95	0.98
K	0.2	0.5	0.8
T_{end}	1	2	5
M	1	2	5

Table 5. Optimal values of hybrid augmented ant colony optimization parameters.

Parameters	Optimal values
Total iteration of improved max–min ant system	200
Number of ants	15
A	2
B	5
ρ	0.2
δ	0.75
M	5
T_0	200
α	0.98
K	0.5
T_{end}	2

HA-ACO evaluation

To evaluate the performance of HA-ACO, 12 instance problems in small and medium sizes and 23 benchmarks in large sizes are solved. Table 6 represents the scale of 12 instance problems and the obtained computational results.

As can be seen, HA-ACO and IMMAS could generate high-quality results for total costs in comparison with the CPLEX solver of GAMS software where HA-ACO and IMMAS have average gaps of 0.65% and 1.07%, respectively. Moreover, HA-ACO and IMMAS could resolve the problems within acceptable run times with respect to the CPLEX solver such that CPLEX, HA-ACO and IMMAS have average run times of 1084.27, 158.37 and 122.10 seconds, respectively. Therefore, we can conclude that HA-ACO demonstrates an acceptable performance to solve small- and medium-sized problems. In order to assess the performance of HA-ACO in large-sized problems, 23 problem instances of gdb files (Belenguer, 1997) are chosen to be solved by HA-ACO and compared with IMMAS. The obtained results are given in Table 7. It should be noted that since the number of vehicles used in the problems are similarly determined by each solution method, the total cost reported in Table 7 only includes the total traversing cost.

In Table 7, from left to right, the columns denote the problem number, the disposal facility's node number, and the maximum available usage time for vehicles, respectively. It is demonstrated that HA-ACO could perform better in all large-sized problems as well as small- and medium-sized problems. IMMAS could yield an average gap of 0.53% compared to HA-ACO, and the average run time of 196.80 seconds while HA-ACO could treat the problems within the average run time of 239.02 seconds. Moreover, it is clarified that the initial values of pheromones have a great impact on the quality of the final solution which are determined efficiently using the SA algorithm. According to the results of Tables 6 and 7, we can conclude that HA-ACO has shown its superiority in all problems that can be implemented as an efficient solution tool in different real-life problems. For this reason, a case study is analyzed in Iran to test the applicability of the suggested methodology of this research.

Case study

This section conducts a real case study in Iran to assess the applicability of the suggested methodology under uncertain conditions. To this end, one of the areas in the southwest of Isfahan city is selected. Currently, waste collection is performed every second night using one vehicle in this area, but its main streets are served every night. The capacity of this vehicle is three tons and it has a maximum available usage time of five hours. Management of the municipality has decided to change the service procedure and to eliminate Friday nights from the service planning (it is worthy to note that Friday is weekend in Iran). This change creates uncertainty on the information of the amount of the generated waste to be collected. This study aims to determine a set of routes to minimize the total traversed distance and the number of vehicles considering fuzzy demands. Figure 5 depicts the satellite map of the given zone in Isfahan.

After surveying Figure 5 and with respect to the initial information, the related graph network of the specified area is extracted which is given in Figure 5. In this network, 99 nodes and 152 edges with 115 required edges are defined. The node number 0 indicates the depot that is not depicted in Figure 6 due to its location which is out of the operational zone. Furthermore, the disposal facility (node number 100) is located outside of the metropolitan area which is 12.3 km away from the operational zone. The other information and the value of fuzzy demands are estimated by the experts' judgments in the municipality. Based upon the experts' opinions, it is expected to have more waste to

Table 6. The comparison results obtained by hybrid augmented ant colony optimization (HA-ACO) for small- and medium-sized problems.

Problem	|V|	|E|	|ER|	|K|	CPLEX		HA-ACO			Improved max–min ant system		
					Total cost ($)	Run time (s)	Total cost ($)	Run time (s)	Gap (%)	Total cost ($)	Run time (s)	Gap (%)
P1	4	5	4	2	230.12	2.5	230.12	21.2	0.00	230.12	18.8	0.00
P2	5	8	6	3	243.9	11.5	243.9	39.6	0.00	243.90	27.7	0.00
P3	6	10	8	4	367.45	26.9	367.45	54.0	0.00	367.45	39.1	0.00
P4	7	15	11	5	379.12	74.7	379.12	84.6	0.00	379.12	65.2	0.00
P5	8	18	11	6	501.98	158.6	504.11	101.2	0.42	506.54	85.7	0.91
P6	9	22	15	8	633.51	593.4	642.33	139.9	1.39	645.10	105.3	1.83
P7	10	25	18	8	756.19	869.2	765.66	151.9	1.25	767.27	121.3	1.47
P8	11	28	21	9	885.6	1020.6	888.63	190.3	0.34	893.01	136.8	0.84
P9	12	30	24	9	1034.28	1485.3	1046.34	219.6	1.17	1058.73	155.5	2.36
P10	13	32	28	10	1193	2110.5	1202.24	251.9	0.77	1211.45	191.8	1.55
P11	14	34	31	11	1226.81	3058	1238.69	287.2	0.97	1245.29	237.5	1.51
P12	15	43	34	12	1391.4	3600	1411.85	359.0	1.47	1424.67	280.5	2.39
Average	–	–	–	–	736.95	1084.27	743.37	158.37	0.65	747.72	122.10	1.07

Table 7. The comparison results obtained by hybrid augmented ant colony optimization (HA-ACO) for large-sized problems.

P/DF/T_{max}	HA-ACO		Improved max–min ant system		
	Total cost ($)	Run time (s)	Total cost ($)	Run time (s)	Gap (%)
1/10/151	494	130.26	498	106.33	0.83
2/10/145.5	527	123.38	528	134.16	0.13
3/10/139.8	480	148.14	480	138.19	0.04
4/10/178.5	456	159.34	461	158.12	1.04
5/10/157.9	545	227.01	549	199.22	0.68
6/10/156	463	193.67	466	181.21	0.74
7/10/157.2	471	181.30	474	173.95	0.72
10/15/63	461	354.77	464	335.29	0.58
11/15/65.7	442	356.55	444	323.01	0.56
12/10/189	321	201.51	324	181.32	0.81
13/10/213.6	511	424.53	512	584.37	0.23
14/5/144	663	219.57	667	229.41	0.61
15/8/254.4	749	263.48	750	236.85	0.12
16/4/57.6	127	210.11	128	191.55	0.56
17/7/42	60	212.88	60	173.42	0.00
18/5/71.4	174	236.56	175	161.40	0.71
19/8/50.4	91	184.20	91	148.10	0.49
20/8/94.8	212	200.66	214	263.32	0.85
21/2/45	78	149.04	78	152.07	0.00
22/8/78.75	186	251.72	187	186.57	0.40
23/5/74.5	219	254.92	220	250.18	0.26
24/2/71.625	266	352.33	268	308.94	0.79
25/3/66.9	302	461.55	305	510.21	0.94
Average	360.80	239.02	362.74	196.80	0.53

be collected on Saturday, and the average amount of waste is considered to be the left boundary of the triangular fuzzy value of each required edge. Moreover, the value of the middle boundary is 1.2 times larger than the value of the left boundary and the value of the right boundary is 1.2 times larger than the value of the middle boundary.

Optimal credibility level

In order to find the optimal credibility level, the problem is solved for different credibility levels. For this reason, HA-ACO is utilized. We consider two possible heterogenous vehicles with the capacities of three tons and two tons, maximum available



Figure 5. The satellite map and its specified area in the case study problem (OpenStreetMap, 2019).

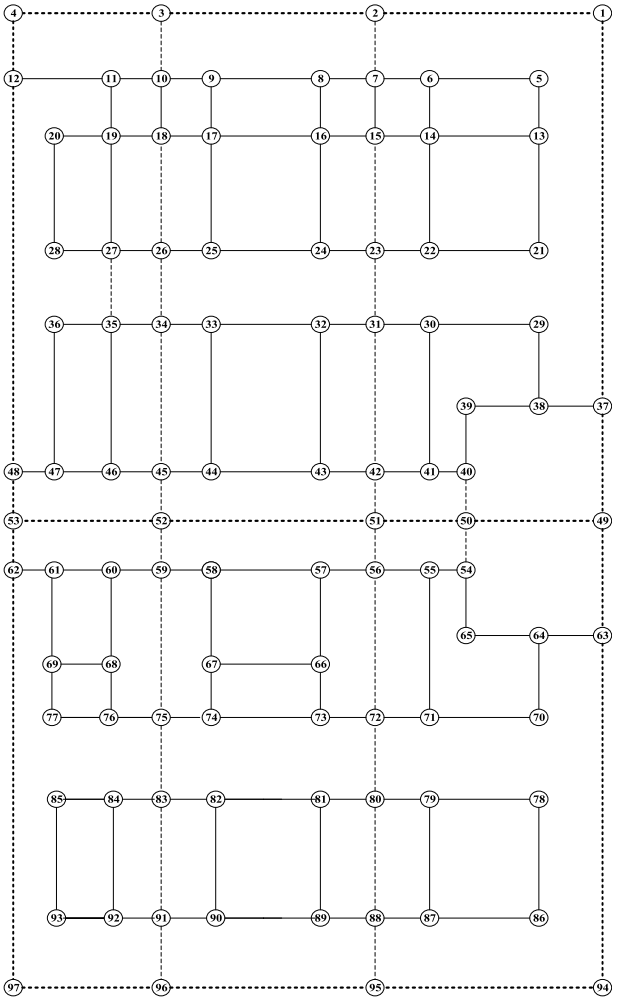


Figure 6. The corresponding graph of the specified area in the case study problem.

time of five hours, fixed usage cost of 1,000,000 units (for three-tons vehicle) and 800,000 units (for two-tons vehicle) and variable usage cost of 140,000 units per hour, respectively. Each vehicle has two crew workers and one driver which impose the total cost of 90,000 units. Vehicles' average speeds are 10 km/hour in the operational area and 80 km/hour outside of the operational area. The traversing time on $e(i, j)$ is determined using the equation of $t_{ij} = x_{ij} / v_{ij}$, where x_{ij} and v_{ij} represent the distance and vehicle' speed on $e(i, j)$. The obtained results for different credibility levels are presented in Table 8.

The additional costs due to route failure are calculated by the proposed simulation algorithm in the "Deremining the optimal credibility level" subsection, then the planned costs and total cost are obtained by the HA-ACO algorithm. Finally, the impact of the dispatcher preference index (Cr) is analyzed using these cost elements. For this purpose, Cr takes different values in the interval [0, 1] by steps of length 0.1. The proposed solution methodology is run for each to obtain the optimal dispatcher preference index (Cr^*).

Table 8. The obtained result of the case study problem.

Parts of object	Deterministic problems	Dispatcher preference index										
		0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
PT1	–	2.4	2.5	2.61	2.76	2.76	3.12	3.38	2.01	3.17	2.95	2.84
PT2		0	0	0	0	0	0	0	1.82	1.16	1.38	1.73
AT1	–	2.32	2.12	1.86	1.57	1.37	1.18	0.67	0	0	0	0
AT2	–	0	0	0	0	0	0	0	0	0	0	0
TT	2.73	4.72	4.62	4.47	4.33	4.13	4.3	4.05	3.83	4.33	4.33	4.57
UV1	1	1	1	1	1	1	1	1	1	1	1	1
UV2	0	0	0	0	0	0	0	0	1	1	1	1
FCost	1245700	1424800	1415800	1402300	1389700	1371700	1387000	1364500	2144700	2189700	2189700	2211300
VCost	382200	660800	646800	625800	606200	578200	602000	567000	536200	606200	606200	639800
Total cost	1627900	2085600	2062600	2028100	1995900	1949900	1989000	1931500	2680900	2795900	2795900	2851100

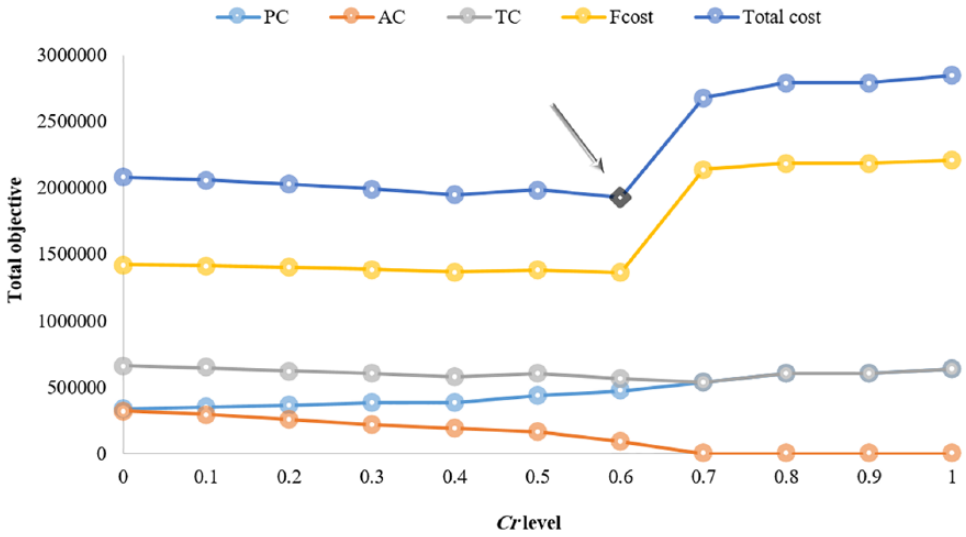


Figure 7. Fluctuations of assessed costs against different dispatcher preference indexes (*Cr*).

Table 9. The routing results of the vehicles in the case study problem.

Vehicles	Trip number	Routing
Three-tons	1	0 - 4 - 12 - 11 - 10 - 9 - 8 - 7 - 6 - 5 - 13 - 14 - 6 - 14 - 15 - 7 - 15 - 16 - 8 -16 - 17 - 9 - 17 - 18 - 10 - 18 - 19 - 11 - 19 - 20 - 28 - 27 - 19 - 27 - 26 - 25 - 17 - 16 - 24 - 25 - 24 - 23 - 22 - 14 - 13 - 21 - 22 - 23 - 31 - 30 - 29 - 38 - 39 - 40 - 41 - 40 - 50 - 49 - 63 - 94 - 100
	2	100 - 49 - 50 - 40 - 41 - 42 - 43 - 32 - 31 - 30 - 41 - 42 - 42 - 43 - 44 - 33 - 32 - 33 - 34 - 35 - 46 - 35 - 36 - 47 - 46 - 45 - 44 - 45 - 46 - 47 - 48 - 53 - 62 - 61 - 69 - 68 - 60 - 61 - 62 - 97 - 100
Two-tons	1	0 - 63 - 64 - 70 - 71 - 55 - 54 - 65 - 64 - 65 - 54 - 55 - 56 - 57 - 66 - 73 - 72 - 71 - 72 - 80 - 79 - 78 - 86 - 87 - 79 - 80 - 88 - 95 - 100
	2	100 - 96 - 91 - 92 - 93 - 85 - 84 - 92 - 91 - 83 - 82 - 90 - 91 - 90 - 89 - 88 - 89 - 81 - 82 - 83 - 84 - 83 - 75 - 74 - 73 - 66 - 67 - 58 - 57 - 58 - 59 - 60 - 68 - 76 - 75 - 76 - 77 - 69 - 61 - 62 - 97 - 100

In Table 8, PT_1, PT_2, AT_1, AT_2, TT, UV_1, UV_2, $FCost$ and $Vcost$ are the objective function values in terms of the planned service time of the three-tons vehicle, the planned service time of the two-tons vehicle, and the additional service time caused by failure in routes of the three-tons vehicle. The additional service time is caused by a failure in routes of the two-tons vehicle, total service time (which is equal to the sum of PT_1, PT_2, AT_1 and AT_2), number of the used three-tons vehicle, number of the used two-tons vehicle, vehicles' fixed costs including vehicles usage cost, driver and crew's wages and variable costs imposed by total service time, respectively. Finally, total cost is the sum of the fixed and variable costs for each given value of Cr. The second column denotes the deterministic problems information, namely, DP. Equation (26) represents the calculations of the above notations

$$\begin{aligned} &FCost = (UV_1 \times 1000000) + (UV_2 \times 800000) + (90000 \times TT), \\ &VCost = TT \times 140000, \\ &\text{Total cost} = VCost + FCost. \end{aligned} \tag{26}$$

It is worth noting that the values of PC and AC as the total planned cost and the total additional cost are calculated based on PT_1, PT_2, AT_1 and AT_2. As an example, for $Cr = 0$, we have

$$\begin{aligned} &PC = PT_1 \times 140000\,\text{units/hr} \\ &= 2.4\,\text{hr} \times 140000\,\text{units/hr} = 336000\ \text{units}, \\ &AC = PT_1 \times\times 140000\,\text{units/hr} \\ &= 2.32\,\text{hr} \times 140000\,\text{units/hr} = 324800\ \text{units}, \\ &VCost = PC + AC \times 660800. \end{aligned} \tag{27}$$

It is obvious from Table 8 that the planned cost follows a rising trend when Cr increases. It is exactly what happens to the additional cost in reverse. The reason is that the vehicles must traverse additional distance due to the failures. According to Figure 7, the optimal value of credibility level (Cr^*), that is, the optimal preference index of the dispatcher is 0.6 and the number of used vehicles to cover all the required edges is also equal to one. Figure 7 shows the fluctuations of different costs by changing Cr. The values of PC, AC, TC, $Fcost$ and total cost are calculated as 473200, 93800, 567000, 1364500 and 1931500 for $Cr^* = 0.6$, respectively.

Now, the routing results of both vehicles are shown in Figure 8 and Table 9. In Figure 8, the red and green lines represent the routes of three-tons and two-tons vehicles, respectively.

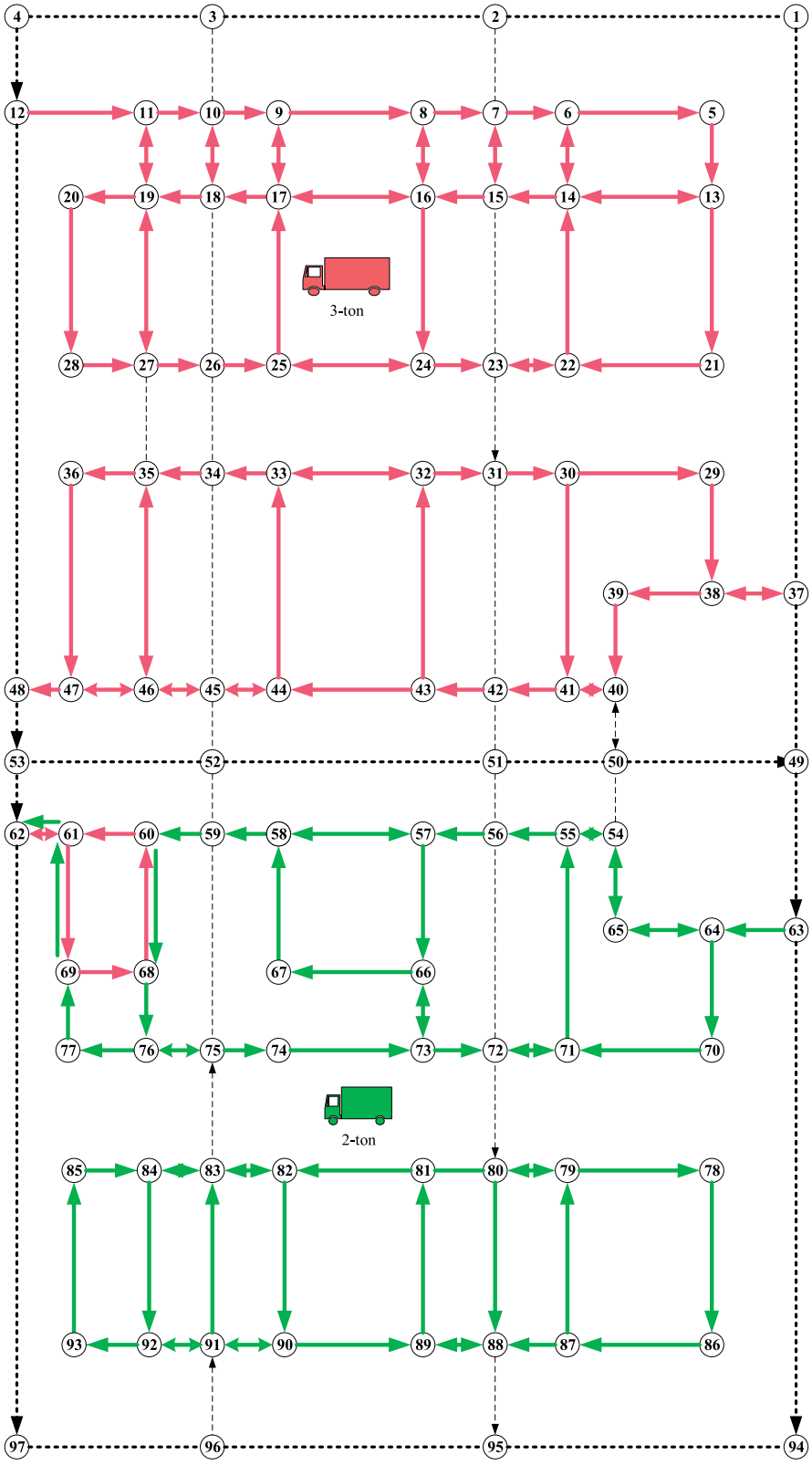


Figure 8. The routing map of the vehicles in the case study problem.

Sensitivity analysis

Sensitivity analysis can be employed as a very useful managerial tool to evaluate different policies under different conditions. In this research, a sensitivity analysis is performed with respect to the maximum available usage time of vehicles (T_{max}) to evaluate the effects of different T_{max} on the behavior of the objective function and the quality of the solution. Therefore, we applied the changes of the maximum available usage time from five hours to seven hours with 15-minutes increments in each change interval.

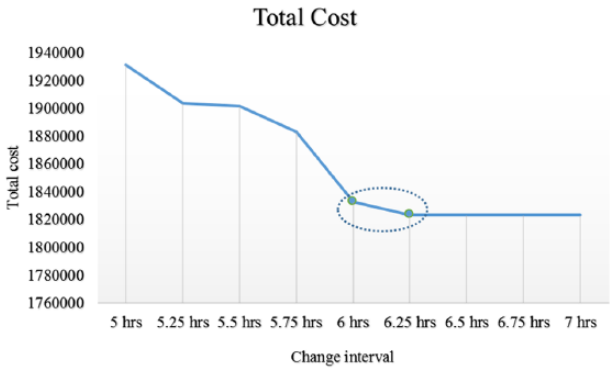


Figure 9. The behavior of the objective function against different values of T_{max} for $Cr^* = 0.6$.

The output analysis results are given in Figure 9 for the optimal credibility value $Cr^* = 0.6$.

As it is obvious, the highest total cost reduction has occurred between the maximum available usage times of five hours and 45 minutes and six hours, which is equal to 50,600 units, such that this may create a critical change in the decision-making process. Moreover, after the interval of $T_{max} = 6.25$ hours, the objective function has shown no change due to creating additional service time for the vehicle when it has already covered all the required edges. As a significant enhancement, we can realize that the additional costs increase by taking a lower value for T_{max}; this leads to moving back to the disposal facility, which is 12.3 km away from the operational area. Hence, the management can utilize this efficient tool to define the optimal policy.

Conclusion

Among the issues raised in the context of USWM, the uncertainty in the amount of generated waste is highly critical, which may change the periodic collection planning of the waste-collection vehicles. This paper focused on the fuzzy multi-trip CARP, aiming at the minimization of the total traversing cost and the total number of vehicles under uncertain conditions. In the proposed problem, the demands of required edges are regarded to be a triangular fuzzy number and the locations of the disposal facility and vehicles' depot are assumed to be separated. The suggested approach to treat uncertainty is fuzzy chance-constrained programming based on the credibility theory and Monte Carlo simulation. To solve the problem, a HA-ACO was developed including a heuristic algorithm, SA algorithm to generate high-quality initial solutions and an IMMAS algorithm to construct the optimal routes. Furthermore, an innovative probability function was defined to enhance the suggested IMMAS and the parameters of HA-ACO were identified optimally using the Taguchi design method. The performance of the suggested HA-ACO was assessed compared to a similar algorithm in the literature, and it is demonstrated that HA-ACO could perform better to solve different instance problems and benchmarks. Finally, a real case study from Iran was conducted with an efficient sensitivity analysis to study the optimal policy and present subsequent improvement suggestions.

For our future studies, locational decisions can be addressed in the problem to make it closer to real-life. The proposed model may be extended by taking into account other objectives such as the minimization of environmental pollution arising from late services and service reliability maximization. Other well-known metaheuristic algorithms such as the genetic algorithm, variable neighborhood search and ant lion optimization algorithm can be implemented and compared with HA-ACO. Moreover, further uncertainty approaches such as grey systems (Roy et al., 2017), robust optimization (Goli et al., 2019; Özmen et al., 2011; Sangaiah et al., 2019; Wang et al., 2018) and stochastic optimal control (Temoçin and Weber, 2014) may be applied to the model.

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